Review 2: Methodology & Implementation Details

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1. Process Lifecycle & State Machine

Patients progress through five formal states: NEW (registered, priority classified), READY (queued in scheduler, accumulating aging ticks), RUNNING (allocated resources, undergoing active medical treatment), WAITING (blocked due to resource contention or Banker's safety check), and COMPLETED (treatment finished, resources returned).

Preemption allows a running process to be moved back to READY if a higher-urgency Level 1 patient arrives. Contested resources are reclaimed, and the preempted task retains its executed burst progress.

2. Mathematical and Algorithmic Formulations

• Dynamic Aging: Effective Priority Peff(t) = max(2, Pbase - floor(t_ready / Ta)), with Ta = 4 ticks. Capping at Level 2 prevents starvation of lower urgencies while ensuring Level 1 Critical cases remain strictly dominant.

• Banker's Avoidance: Maintained via Available vector, Max matrix, Allocation matrix, and Need = Max - Allocation. Before granting any request Req <= Need, safety state is verified using Work and Finish vectors.

• Deadlock Detection (WFG): Directed edge Pi -> Pj forms if Pi is waiting for resource R held by Pj when Available[R] == 0. A recursive DFS flags cycles.

• Preemption Recovery: The cycle victim is chosen by min(-Peff, Progress, -ArrivalTime). The victim is preempted to READY, releasing resources to clear circular wait.

3. Technology Stack & Software Architecture (C Implementation)

Layer	Technology	Description
Core Engine	C Language (C11 Standard)	Low-level system programming, memory-efficient process execution
Concurrency	POSIX Threads (pthreads)	Multi-threaded synchronization using pthread_mutex_t and pthread_cond_t
Synchronization	Counting Semaphores & Mutex	Guards 5 resource pools and guarantees atomic allocation critical sections
Build System	GNU Make (Makefile)	Compiles herm_cli and herm_server with clang -Wall -Wextra -O2 -pthread
Networking	BSD Sockets (sys/socket.h)	Pure C embedded HTTP server serving REST API on port 5050
Command Center	HTML5, CSS3, ES6 JavaScript	Aesthetic dark-mode clinical command center with live Kanban
Visual Graph	SVG Vector Canvas	Interactive Wait-For Graph with dynamic node layout and cycle arrows

4. Initial Test Workload (Proposal Validation)

Patient	Arrival	Priority	Burst	Resource Demands
P1	0	3 (Moderate)	8	1 Bed, 1 Doctor, 2 Nurses
P2	1	2 (Serious)	6	1 Bed, 1 Ventilator, 1 Doctor

P3	2	4 (Stable)	4	1 Doctor, 1 Nurse (Round Robin, Aging eligible)
P4	3	1 (Critical)	5	1 Bed, 1 Ventilator, 2 Doctors, 2 Nurses, 1 Theatre
P5	4	3 (Moderate)	7	1 Bed, 1 Doctor, 1 Nurse
P6	6	1 (Critical)	4	1 Bed, 1 Ventilator, 1 Doctor, 1 Theatre